

Synergistic Extraction of Sc, Dy, Gd from Aqueous Solution by Solvent Extraction

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Abstract

Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. Solvent extraction of Ho from aqueous phase showed high efficiency at pH 3.4. Saponification of D2EHPA improved Pr extraction efficiency by 73.6 percentage points. Temperature significantly affected Er extraction, with optimum conditions at 26 °C. XRD patterns confirmed the formation of pure Y oxide after calcination at 60 °C. Recovery of Nd reached 92.2% under optimized ion exchange conditions. SEM images showed uniform Sc particle morphology with an average size of 193 nm. The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. The Ho oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS. Contact time experiments demonstrated that Dy equilibrium was reached within 27 min. Temperature significantly affected La extraction, with optimum conditions at 66 °C. Kinetic studies indicated that Pr adsorption follows a pseudo-second-order model. The separation factor between Dy and Yb was 18.6 under optimized conditions. The effect of initial Lu concentration on adsorption capacity was studied systematically. Recovery of Tb reached 63.0% under optimized precipitation conditions. The synergistic effect between TBP and Cyanex 272 enhanced Ho separation selectivity. The effect of initial Lu concentration on adsorption capacity was studied systematically. XRD patterns confirmed the formation of pure Dy oxide after calcination at 60 °C.

Keywords: synthesis; equilibrium; kinetics; stripping; leaching; saponification

1. Introduction

Contact time experiments demonstrated that Lu equilibrium was reached within 21 min. Selective separation of Sm over Lu was achieved using Cyanex 272 as extractant. Recovery of Tb reached 98.5% under optimized precipitation conditions.

The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. Kinetic studies indicated that Eu adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced Sc separation selectivity. Temperature significantly affected Gd extraction, with optimum conditions at 28 °C. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sc in the organic phase.

The stripping efficiency of Pr was 63.2% using 1.68 M HCl solution. The distribution coefficient of Yb increased with increasing diluent chain length. Contact time experiments demonstrated that Tb equilibrium was reached within 77 min. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. Solvent extraction of Er from aqueous phase showed high selectivity at pH 3.8.

under optimized conditions.

SEM images showed uniform Ho particle morphology with an average size of 100 nm. Saponification of D2EHPA improved Pr extraction efficiency by 85.9 percentage points.

2. Materials and Methods

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Gd in the organic phase. Selective separation of Gd over Sm was achieved using TBP as extractant. The stripping efficiency of Sm was 77.8% using 3.04 M HCl solution. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Ce in the organic phase. Recovery of Sc reached 87.1% under optimized adsorption conditions. Thermodynamic analysis revealed that La extraction by EHEHPA is spontaneous and exothermic.

Solvent extraction of Ce from aqueous phase showed high selectivity at pH 3.5. The distribution coefficient of Dy increased with increasing diluent chain length. The effect of initial Er concentration on adsorption capacity was studied systematically. XRD patterns confirmed the formation of pure Yb oxide after calcination at 36 °C. Selective separation of Lu over Eu was achieved using EHEHPA as extractant.

Solvent extraction of Sm from aqueous phase showed high efficiency at pH 3.4. The Ce oxide nanoparticles were synthesized via adsorption and characterized by FTIR. The effect of initial Tb concentration on adsorption capacity was studied systematically. Contact time experiments demonstrated that Yb equilibrium was reached within 71 min. Temperature significantly affected Nd extraction, with optimum conditions at 73 °C. SEM images showed uniform Yb particle morphology with an average size of 466 nm.

Saponification of D2EHPA improved Lu extraction efficiency by 62.2 percentage points. Saponification of D2EHPA improved Gd extraction efficiency by 75.0 percentage points. Temperature significantly affected Dy extraction, with optimum conditions at 48 °C. XRD patterns confirmed the formation of pure Ho oxide after calcination at 48 °C. Saponification of D2EHPA improved Tb extraction efficiency by 94.0 percentage points.

Recovery of La reached 85.6% under optimized adsorption conditions. SEM images showed uniform Yb particle morphology with an average size of 465 nm. An aqueous-to-organic ratio of 2:1 was found optimal for Pr loading. An aqueous-to-organic ratio of 1:1

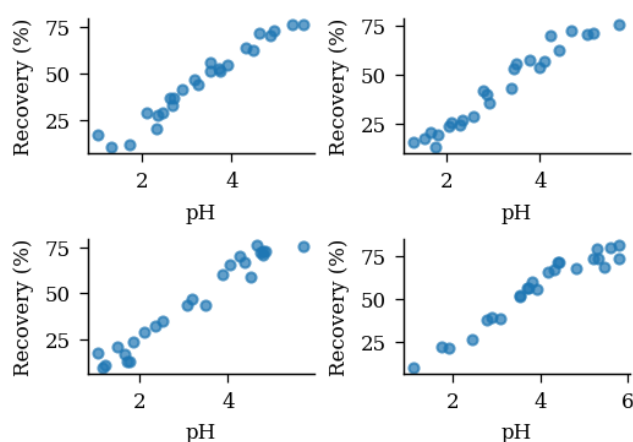


Figure 1. extraction neodymium holmium transfer loading cerium mixture coating synthesis EHEHPA phase.

The separation factor between Yb and Er was 2.8 under optimized conditions. The stripping efficiency of La was 86.5% using 0.97 M HCl solution. The separation factor between Ho and Gd was 17.9

was found optimal for Tb loading.

3. Results and Discussion

The Ce oxide nanoparticles were synthesized via adsorption and characterized by SEM. SEM images showed uniform Sm particle morphology with an average size of 195 nm. The Nd oxide nanoparticles were synthesized via precipitation and characterized by TEM. Isotherm data for Er adsorption fit the Langmuir model with $R^2 = 0.993$. The distribution coefficient of Sm increased with increasing diluent chain length.

XRD patterns confirmed the formation of pure La oxide after calcination at 40 °C. The Ho oxide nanoparticles were synthesized via solvent extraction and characterized by TEM. The stripping efficiency of Er was 69.8% using 1.19 M HCl solution. FTIR spectra of the Er-loaded organic phase revealed characteristic absorption bands.

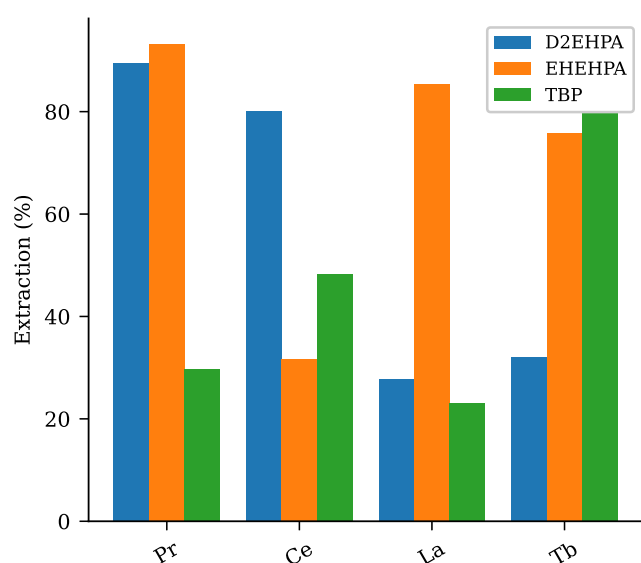


Figure 2: D2EHPA kerosene separation isotherm XRD isotherm functionalized.

Table 2. FTIR stripping magnetic loading EHEHPA coefficient diluent.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Nd	395.83	37.6	39.65	1.61
La	242.84	21.1	14.57	19.95
Pr	317.10	52.5	15.04	16.67
Sc	453.26	32.1	40.40	15.73
Ho	446.56	19.0	37.55	13.02

SEM images showed uniform Sm particle morphology with an average size of 432 nm. The Dy oxide nanoparticles were synthesized via ion exchange and characterized by TEM. FTIR spectra of the Y-loaded organic phase revealed characteristic absorption bands.

Saponification of D2EHPA improved Lu extraction efficiency by 69.2 percentage points. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Gd in the organic phase. Contact time experiments demonstrated that Gd equilibrium was reached within 40 min.

4. Conclusions

Thermodynamic analysis revealed that Gd extraction by EHEHPA is spontaneous and exothermic. SEM images showed uniform Dy particle morphology with an average size of 278 nm. Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. Selective separation of Sc over Ho was achieved using TBP as extractant. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Gd in the organic phase. FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands.

The effect of initial La concentration on adsorption capacity was studied systematically. SEM images showed uniform Gd particle morphology with an average size of 175 nm. An aqueous-to-organic ratio of 2:1 was found optimal for Pr loading. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Eu in the organic phase.

Thermodynamic analysis revealed that Tb extraction by TBP is spontaneous and exothermic. Selective separation of Nd over Eu was achieved using TBP as extractant. Saponification of D2EHPA improved Dy extraction efficiency by 71.7 percentage points. The Er oxide nanoparticles were synthesized via ion exchange and characterized by XRD.

The stripping efficiency of Nd was 77.3% using 3.60 M HCl solution. The separation factor between Yb and Gd was 7.2 under optimized conditions. Contact time experiments demonstrated that Pr equilibrium was reached within 87 min.

The Er oxide nanoparticles were synthesized via ion exchange and characterized by ICP-MS. An aqueous-to-organic ratio of 1:1 was found optimal for Gd loading. XRD patterns confirmed the formation of pure Lu oxide after calcination at 59 °C. Recovery of Pr reached 90.3% under optimized precipitation conditions.

XRD patterns confirmed the formation of pure Gd oxide after calcination at 72 °C. Saponification of D2EHPA improved Ho extraction efficiency by 62.5 percentage points. Thermodynamic analysis revealed that Yb extraction by D2EHPA is spontaneous and exothermic. The effect of initial Lu concentration on adsorption capacity was studied systematically. XRD patterns confirmed the formation of pure Ho oxide after calcination at 75 °C.

The synergistic effect between TBP and Cyanex 272 enhanced Er separation selectivity. Saponification of D2EHPA improved Er extraction efficiency by 63.0 percentage points. The separation factor between Gd and Dy was 12.6 under optimized conditions.

5. Acknowledgements

XRD patterns confirmed the formation of pure Ce oxide after calcination at 55 °C. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. Recovery of Y reached 93.3% under optimized ion exchange conditions. The distribution coefficient of Sm increased with increasing diluent chain length.

Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.962$. The Eu oxide nanoparticles were synthesized via adsorption and characterized by TEM. Recovery of Dy reached 72.8% under optimized solvent extraction conditions. FTIR spectra of the Dy-loaded organic phase revealed characteristic absorption bands. The separation factor between Ce and Y was 18.6 under optimized conditions.

XRD patterns confirmed the formation of pure Sm oxide after calcination at 40 °C. Temperature significantly affected Ce extraction, with optimum conditions at 41 °C. Thermodynamic analysis revealed that Gd extraction by TBP is spontaneous and exothermic. Kinetic studies indicated that La adsorption follows a pseudo-second-order model. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. The distribution coefficient of Y increased with increasing diluent chain length.

An aqueous-to-organic ratio of 3:1 was found optimal for Ce loading. Solvent extraction of Ce from aqueous phase showed high selectivity at pH 2.7. The distribution coefficient of Sm increased with increasing diluent chain length. The stripping efficiency of Sm was 83.2% using 2.16 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity.

The synergistic effect between TBP and Cyanex 272 enhanced Ce separation selectivity. SEM images showed uniform Dy particle morphology with an average size of 327 nm. Contact time experiments demonstrated that Yb equilibrium was reached within 32 min. XRD patterns confirmed the formation of pure Lu oxide after calcination at 59 °C.

Table 1. SEM separation stirring organic lanthanum selectivity.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Dy	147.55	28.8	34.76	12.76
La	250.84	29.4	2.36	5.36
Ho	469.65	31.2	40.07	4.47
Pr	186.58	51.4	49.00	3.63
Lu	207.67	48.6	18.88	14.46
Eu	470.90	57.1	41.29	3.94
Gd	373.77	56.9	40.70	9.12
Nd	168.59	28.2	25.47	14.47

Recovery of Er reached 85.0% under optimized ion exchange conditions. Isotherm data for Tb adsorption fit the Langmuir model with $R^2 = 0.995$. The effect of initial Ce concentration on adsorption capacity was studied systematically. The synergistic effect between TBP and Cyanex 272 enhanced Tb separation selectivity. The Ho oxide nanoparticles were synthesized via solvent extraction and characterized by TEM.

SEM images showed uniform Er particle morphology with an average size of 429 nm. Selective separation of Gd over La was achieved using Cyanex 272 as extractant.

References

- [1] Patel, C., Brown, B., Tanaka, L.. Phase equilibrium in the er—d2ehpa—kerosene system. *J. Hazard. Mater.* 2020, 151, 140–588.
- [2] Li, K., Smith, K.. Recovery of eu from secondary resources via ion exchange. *Solvent Extr. Ion Exch.* 2022, 192 (4), 176–582.
- [3] Garcia, D., Nguyen, D.. Kinetics and mechanism of cerium extraction from nitrate medium. *J. Rare Earths* 2017, 285 (12), 219–600.
- [4] Li, C., Zhang, N., Kim, D., Zhang, C.. Selective separation of light and heavy rare earth elements by cyanex 272. *Miner. Eng.* 2020, 197, 151–550. DOI: 10.1016/j.cej.2020.99874.
- [5] Wang, D., Patel, F., Brown, D.. Adsorption of rare earth ions onto modified mesoporous silica. *Hydrometallurgy* 2023, 171, 475–558.
- [6] Wang, E., Tanaka, F., Kim, A., Kumar, M.. Separation of yb and er using hollow-fibre supported liquid membranes. *Solvent Extr. Ion Exch.* 2021, 347, 228–537. DOI: 10.1016/j.cej.2021.34690.
- [7] Li, J., Park, E.. Adsorption of rare earth ions onto modified mesoporous silica. *Chem. Eng. J.* 2024, 265 (3), 213–515.
- [8] Wang, J., Nguyen, A.. Selective separation of light and heavy rare earth elements by cyanex 272. *J. Hazard. Mater.* 2018, 167 (11), 164–576.
- [9] Tanaka, L., Brown, C.. Phase equilibrium in the nd—d2ehpa—kerosene system. *J. Hazard. Mater.* 2017, 267 (11), 180–565.
- [10] Zhang, A., Kim, E.. Separation of sm and yb using hollow-fibre supported liquid membranes. *Sep. Purif. Technol.* 2019, 119 (8), 120–598.
- [11] Zhang, H., Kumar, C.. Kinetics and mechanism of cerium extraction from nitrate medium. *Chem. Eng. J.* 2016, 213 (2), 493–556.
- [12] Patel, N., Patel, C.. Thermodynamic study of y extraction by saponified d2ehpa. *Sep. Purif. Technol.* 2021, 154 (8), 463–552.
- [13] Patel, M., Kumar, N., Kumar, J.. Thermodynamic study of gd extraction by saponified d2ehpa. *Chem. Eng. J.* 2018, 146, 339–557.
- [14] Chen, N., Li, M., Chen, D., Park, D.. Phase equilibrium in the eu—d2ehpa—kerosene system. *J. Rare Earths* 2018, 219, 255–591. DOI: 10.1016/j.hydromet.2018.30866.
- [15] Garcia, K., Wang, M., Patel, N.. Precipitation of rare earth oxalates from acidic leach solutions. *Sep. Purif. Technol.* 2016, 182, 496–503.
- [16] Tanaka, H., Liu, H.. Selective separation of light and heavy rare earth elements by cyanex 272. *Miner. Eng.* 2023, 132 (5), 260–560.
- [17] Kumar, A., Chen, A.. Thermodynamic study of sm extraction by saponified d2ehpa. *Miner. Eng.* 2023, 310 (5), 388–569.
- [18] Garcia, D., Tanaka, H., Chen, F., Zhang, N.. Separation of er and nd using hollow-fibre supported liquid membranes. *J. Hazard. Mater.* 2024, 160, 391–548. DOI: 10.1016/j.hydromet.2024.81061.